

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416184
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Niegmann; Kaj

Coupling arrangement for lock device, and lock device

Abstract

Coupling arrangement for a lock device, the coupling arrangement comprising an input and an output element rotatable about a rotation axis; an engaging member movable between a first and second position; an electric motor arranged to affect movement of the engaging member between the first and second positions; a coupling member rotationally locked to the input element and axially movable relative to the input element between a decoupled position, where a rotation of the input element is not transmitted by the coupling member to a rotation of the output element, and a coupled position where a rotation of the input element is transmitted by the coupling member to a rotation of the output element; and a transmission mechanism arranged to transmit a rotation of the coupling member to a movement of the coupling member from the decoupled position to the coupled position when the engaging member adopts the second position.

Inventors:	Niegmann; Kaj (Hägersten, SE)
Applicant:	ASSA ABLOY AB (Stockholm, SE)
Family ID:	1000008820841
Assignee:	ASSA ABLOY AB (Stockholm, SE)
Appl. No.:	18/257949
Filed (or PCT Filed):	December 01, 2021
PCT No.:	PCT/EP2021/083806
PCT Pub. No.:	WO2022/128480
PCT Pub. Date:	June 23, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240018803 A1	Jan. 18, 2024

Foreign Application Priority Data

SE

2051507-8

Dec. 18, 2020

Publication Classification

Int. Cl.: E05B47/00 (20060101); E05B13/00 (20060101)

U.S. Cl.:

CPC E05B47/0012 (20130101); E05B13/005 (20130101); E05B2047/0017 (20130101); E05B2047/0023 (20130101)

Field of Classification Search

CPC: E05B (17/044); E05B (17/045); E05B (17/049); E05B (13/005); E05B (13/101); E05B (13/106); E05B (47/0676); E05B (47/068); E05B (47/0684); E05B (47/0692); E05B (2047/0026); E05B (2047/0028); E05B (2047/0031)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6487884	12/2001	Constantinou	70/279.1	E05B 47/068
7770944	12/2009	Yuan	292/195	E05B 47/06
10597902	12/2019	Guo	N/A	E05B 47/068
2005/0050928	12/2004	Frolov	70/277	E05B 47/068
2006/0254328	12/2005	Pearson	N/A	N/A
2017/0306646	12/2016	Guo et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
202008006082	12/2007	DE	N/A
102008028800	12/2008	DE	N/A
1046770	12/1999	EP	N/A
2204518	12/2009	EP	N/A
3741934	12/2019	EP	N/A
4843738	12/2010	JP	N/A
5173529	12/2012	JP	N/A
20200079046	12/2019	KR	N/A
524285	12/2003	SE	N/A
9964704	12/1998	WO	N/A

OTHER PUBLICATIONS

“International Application Serial No. PCT EP2021 083806, International Search Report mailed Mar. 31, 2022”, 5 pgs. cited by applicant

“International Application Serial No. PCT EP2021 083806, Written Opinion mailed Mar. 31, 2022”, 9 pgs. cited by applicant

“Swedish Application Serial No. 2051507-8, Office Action mailed Sep. 6, 2021”, 10 pgs. cited by

applicant

“Swedish Application Serial No. 2051507-8, Office Action mailed Nov. 22, 2022”, 5 pgs. cited by applicant

Primary Examiner: Merlino; Alyson M

Attorney, Agent or Firm: Schwegman Lundberg & Woessner, P.A.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application is a national stage application under 35 U.S.C. § 371 of PCT Appl. No. PCT/EP2021/083806, titled “Coupling Arrangement for Lock Device, and Lock Device,” filed Dec. 1, 2021, which claims priority to Swedish Patent Appl. No. 2051507-8, filed Dec. 18, 2020, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure generally relates to a coupling arrangement for a lock device. In particular, a coupling arrangement for a lock device, which coupling arrangement comprises an input element, an output element and a coupling member, and a lock device comprising such coupling arrangement, are provided.

BACKGROUND

(3) Some lock devices comprise a cylinder housing, an output element rotatable relative to the cylinder housing about a rotation axis, a knob rotatable about the rotation axis, and an electromechanical coupling arrangement for selectively coupling the knob with the output element. When a user has been authorized, the coupling arrangement couples the knob and the output element and the lock device can be opened by turning the knob. Such lock device may be referred to as a digital lock cylinder.

(4) In some lock devices of this type, the coupling arrangement comprises an electric motor, and an engaging member, such as a pin, driven by the electric motor linearly in parallel with the rotation axis between a decoupled position and a coupled position. The arrangement of the electric motor in parallel with the rotation axis and the movement of the engaging member in parallel with the rotation axis however add size to the lock device in the axial direction, i.e. along the rotation axis. The lock device may therefore protrude from a door face and may thereby require an escutcheon or rosette to hide the protrusion.

SUMMARY

(5) One object of the present disclosure is to provide a coupling arrangement for a lock device, which coupling arrangement has a compact design.

(6) A further object of the present disclosure is to provide a coupling arrangement for a lock device, which coupling arrangement has a small size.

(7) A still further object of the present disclosure is to provide a coupling arrangement for a lock device, which coupling arrangement has a cost-effective design.

(8) A still further object of the present disclosure is to provide a coupling arrangement for a lock device, which coupling arrangement has a low power consumption.

(9) A still further object of the present disclosure is to provide a coupling arrangement for a lock device, which coupling arrangement has a design of low complexity.

(10) A still further object of the present disclosure is to provide a coupling arrangement for a lock device, which coupling arrangement is secure.

(11) A still further object of the present disclosure is to provide a coupling arrangement for a lock

device, which coupling arrangement provides a reliable operation.

(12) A still further object of the present disclosure is to provide a coupling arrangement for a lock device, which coupling arrangement solves several or all of the foregoing objects in combination.

(13) A still further object of the present disclosure is to provide a lock device comprising a coupling arrangement, which lock device solves one, several or all of the foregoing objects.

(14) According to one aspect, there is provided a coupling arrangement for a lock device, the coupling arrangement comprising an input element rotatable about a rotation axis; an output element rotatable about the rotation axis; an engaging member movable between a first position and a second position; an electric motor arranged to affect movement of the engaging member between the first position and the second position; a coupling member rotationally locked to the input element with respect to the rotation axis and axially movable relative to the input element along the rotation axis between a decoupled position, where a rotation of the input element is not transmitted by the coupling member to a rotation of the output element, and a coupled position, where a rotation of the input element is transmitted by the coupling member to a rotation of the output element; and a transmission mechanism arranged to transmit a rotation of the coupling member to a movement of the coupling member from the decoupled position to the coupled position when the engaging member adopts the second position.

(15) When the coupling member adopts the coupled position, the output element of the lock device can be rotated by rotating the input element. Thus, in the coupled position of the coupling member, the lock device can be locked or unlocked by rotating the input element.

(16) When the coupling member adopts the decoupled position, a rotation of the input element is not transmitted to a rotation of the output element. In the decoupled position of the coupling member, the input element and the coupling member may be freely rotatable around the rotation axis relative to the output element. Thus, in the decoupled position of the coupling member, the lock device cannot be locked or unlocked by rotating the input element.

(17) The coupling member is arranged to not transmit a rotation of the input element to a rotation of the output element in the decoupled position, and to transmit a rotation of the input element to a rotation of the output element in the coupled position. The coupling member thereby functions as a clutch.

(18) In the second position, the engaging member may engage the coupling member. In the first position, the engaging member may be disengaged from the coupling member. The engaging member may be arranged to adopt any of the first position and the second position when the coupling member is in the decoupled position. The coupling member may comprise one or more recesses. The engaging member may engage one recess in the second position and may be disengaged from each recess in the first position.

(19) Since a rotation of the coupling member is transmitted by the transmission mechanism to a movement of the coupling member from the decoupled position to the coupled position when the engaging member adopts the second position, the coupling arrangement enables the engaging member to move between the first position and the second position in a direction perpendicular to the rotation axis. This in turn enables the coupling arrangement to be made more compact and smaller along the rotation axis, in comparison with a prior art coupling arrangement where an engaging member moves in parallel with the rotation axis between a decoupled position and a coupled position engaging the output element. In this way, also the overall lock device can be made more compact and smaller.

(20) The input element may be fixed to, or constituted by, a manually maneuverable actuating element, such as a knob. Since a rotation of the input element is transmitted to a movement of the coupling member from the decoupled position to the coupled position when the engaging member adopts the second position, the coupling member is moved from the decoupled position to the coupled position by manual force, and not by the electric motor. The position of the engaging member (which is controlled by the electric motor) is however decisive of whether or not the

coupling member will be moved from the decoupled position to the coupled position when the input element is rotated by manual force. The electric motor therefore indirectly contributes to movement of the coupling member from the decoupled position to the coupled position.

(21) Since the coupling member is moved from the decoupled position to the coupled position by rotation of the input element, a distance for the engaging member between the first position and the second position can be rather short. This further contributes to a compact design of the coupling arrangement.

(22) The transmission mechanism may take various forms in order to mechanically transmit a rotation of the coupling member to a movement of the coupling member along the rotation axis from the decoupled position to the coupled position when the engaging member adopts the second position. According to one variant, the transmission mechanism is formed by the coupling member, or by the engaging member and the coupling member. The transmission mechanism may however optionally comprise further components in addition to the engaging member and the coupling member.

(23) The input element may be configured to rotate continuously about the rotation axis. Thereby, a seamless access can be provided.

(24) The coupling member may move in a forward direction when moving from the decoupled position to the coupled position. Conversely, the coupling member may move in a rearward direction when moving from the coupled position to the decoupled position. The output element may be arranged forward of the coupling member.

(25) The coupling member may be slidable along the input element between the decoupled position and the coupled position. The coupling member may thus be a sliding member. The coupling member may enclose the input element.

(26) Alternatively, or in addition, the coupling member may be concentric with the rotation axis.

(27) The input element may comprise one or more input shafts. Correspondingly, the output element may comprise one or more output shafts.

(28) The output element may comprise a locking member, such as a tailpiece. The output element may be arranged to drive a bolt in a lock case between an engaged position where the bolt engages a strike plate, and a disengaged position where the bolt is retracted from the strike plate.

(29) The second position may lie radially inward of the first position with respect to the rotation axis. The coupling member may have a generally cylindrical shape, or cylindrical shape.

(30) The electric motor may be a linear motor for affecting movement of the engaging member between the first position and the second position. The electric motor may be powered by a battery. Alternatively, or in addition, the electric motor may be powered by means of energy harvesting.

(31) The electric motor may be arranged to affect movement of the engaging member between the first position and the second position linearly in an actuating direction. The electric motor may be elongated. In this case, the electric motor may be concentric with the actuating direction.

(32) The actuating direction and the rotation axis may be non-parallel. The actuating direction may for example be angled 45 degrees to 135 degrees, such as 80 degrees to 100 degrees, to the rotation axis.

(33) The actuating direction may be angled approximately 90 degrees, such as 90 degrees, to the rotation axis.

(34) The coupling arrangement may further comprise an actuating member arranged to be driven by the electric motor, and an engaging spring arranged to force the engaging member relative to the actuating member from the first position towards the second position. The actuating member, the engaging spring and the engaging member may form a spring loaded assembly for selectively engaging the coupling member. The engaging spring may be arranged between the actuating member and the engaging member.

(35) The electric motor may comprise a threaded and rotatable motor shaft. In this case, the actuating member may threadingly engage the motor shaft such that a rotation of the motor shaft

causes a linear movement of the actuating member.

(36) The actuating member may be movable between a first actuating position and a second actuating position. In this case, the engaging spring may be arranged to force the engaging member from the first position to the second position when the actuating member adopts the second actuating position. Thus, by moving the actuating member from the first actuating position to the second actuating position, the electric motor can affect movement of the engaging member from the first position to the second position.

(37) For example, when the electric motor is not actuated such that the actuating member is in the first actuating position, the engaging member is held in the first position by the engaging spring. When the electric motor is actuated such that the actuating member moves from the first actuating position to the second actuating position and the engaging member is not aligned with a recess in the coupling member, the actuation of the electric motor causes the engaging spring to be compressed. When the coupling member is rotated such that the engaging member becomes aligned with a recess in the engaging member, the engaging spring forces the engaging member relative to the actuating member from the first position and into the second position where the engaging member enters the recess. The engaging member may thus be arranged to adopt the second position only when the actuating member is in the second actuating position and when one recess of the coupling member is rotationally aligned with the engaging member with respect to the rotation axis.

(38) The transmission mechanism may comprise at least one cam profile. In this case, the engaging member may be arranged to engage one of the at least one cam profile in the second position. Each cam profile may be provided on the coupling member, e.g. on a radially outer circumference thereof. Each cam profile may comprise one or more surfaces at an angle to the rotation axis, e.g. angled 40 degrees to 80 degrees relative to the rotation axis.

(39) Each cam profile may be an edge of a recess in the coupling member. When the engaging member is in the second position and engages one of the one or more cam profiles, the engagement causes the coupling member to be pushed forward along the rotation axis when the coupling member rotates around the rotation axis.

(40) The coupling member may comprise a circular track. In this case, the engaging member may be arranged to engage the track. The track may partly or entirely surround the coupling member. In any case, the track may lie in a plane perpendicular to the rotation axis. When the engaging member engages the track, the coupling member is held in the coupled position. Each cam profile may lie between the track and the output element.

(41) The electric motor may be arranged to affect movement of the engaging member from the second position to a third position. In this case, the second position may lie between the first position and the third position, and the engaging member may be arranged to engage the track when the engaging member adopts the third position and the coupling member adopts the coupled position. Also in the third position, the engaging member may engage the coupling member. The movement of the engaging member from the first position to the second position, and the subsequent movement of the engaging member from the second position to the third position, provide a two step engagement of the coupling member by the engaging member.

(42) The third position may lie radially inward of the second position with respect to the rotation axis. The second position may lie radially between the first position and the third position with respect to the rotation axis.

(43) As mentioned above, the engaging member may be arranged to adopt any of the first position and the second position when the coupling member adopts the decoupled position along the rotation axis. The engaging member may be arranged to adopt the third position when the coupling member adopts the coupled position along the rotation axis. The coupling member may be closer to the output element in the coupled position than in the decoupled position.

(44) As a possible alternative, the engaging member may engage the track while the engaging

member is in the second position. For example, an angular distance between two recesses in the coupling member may be utilized to rotate the output element by means of the input element when the coupling member adopts the coupled position.

(45) The engaging spring may be arranged to force the engaging member from the second position to the third position when the actuating member adopts the second actuating position. Thus, a distance moved by the actuating member from the first actuating position to the second actuating position may be equal to or larger than a distance moved by the engaging member from the first position to the third position.

(46) The coupling arrangement may further comprise a coupling arranged to transmit a rotation of the coupling member to a rotation of the output element when the coupling member adopts the coupled position. The coupling may comprise a first coupling part and a second coupling part. The first coupling part may be provided on the coupling member and/or the second coupling part may be provided on the output element.

(47) The coupling may be a Hirth coupling. A Hirth coupling provides a self-centering effect and thereby prevents jamming of the coupling arrangement. The Hirth coupling may comprise tapered teeth on each of the first coupling part and the second coupling part that mesh together when the coupling member adopts the coupled position. The coupling may alternatively be a dog clutch.

(48) The coupling arrangement may further comprise a coupling force device arranged to force the coupling member towards the decoupled position. The coupling force device may be a spring, such as a compression coil spring. When the engaging member is in the third position, the coupling member is held by the engaging member in the coupled position against the force of the coupling force device. When the engaging member is retracted from the third position to the first position, the coupling force device forces the coupling member from the coupled position back to the decoupled position, e.g. in the rearward direction along the rotation axis.

(49) According to a further aspect, there is provided a lock device comprising a coupling arrangement according to the present disclosure. The lock device may for example be a lock cylinder. In this case, the lock device may constitute a digital lock cylinder or an electromechanical lock cylinder. The lock device can replace various lock cylinders, for example a door lock, a padlock or a bike lock.

(50) The lock device may further comprise a cylinder housing. In this case, the electric motor, the engaging member and the coupling member may be arranged inside the cylinder housing. In this way, resistance against tampering or sabotage can be further prevented.

(51) The lock device may further comprise a manually maneuverable actuating element. The actuating element may for example be a lever handle, a knob or a thumbturn. The actuating element may be configured to be contacted and moved by a hand of a user. The actuating element may be rigidly connected to, or integrally formed with, the input element. When the user turns the actuating element, the input element and the coupling member rotate in common about the rotation axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further details, advantages and aspects of the present disclosure will become apparent from the following description taken in conjunction with the drawings, wherein:

(2) FIG. 1: schematically represents a perspective view of a lock device comprising a coupling arrangement;

(3) FIG. 2: schematically represents a cross-sectional side view of the lock device;

(4) FIG. 3: schematically represents a partial perspective view of the lock device;

(5) FIG. 4: schematically represents a block diagram of the lock device;

- (6) FIG. 5: schematically represents a perspective view of the coupling arrangement when an actuating member is in a first actuating position, an engaging member is in a first position, and a coupling member is in a decoupled position;
- (7) FIG. 6: schematically represents a perspective view of the coupling arrangement when the actuating member is in a second actuating position;
- (8) FIG. 7: schematically represents a perspective view of the coupling arrangement when the engaging member is in a second position;
- (9) FIG. 8: schematically represents a perspective view of the coupling arrangement when an input element is rotated;
- (10) FIG. 9: schematically represents a perspective view of the coupling arrangement when the input element is rotated further, the coupling member is in a coupled position, and the engaging member is in a third position;
- (11) FIG. 10: schematically represents a perspective view of the coupling arrangement when an output element is rotated by rotation of the input member; and
- (12) FIG. 11: schematically represents a perspective view of the coupling arrangement when the actuating member has returned to the first actuating position, the engaging member has returned to the first position, and the coupling member has returned to the decoupled position.

DETAILED DESCRIPTION

- (13) In the following, a coupling arrangement for a lock device, which coupling arrangement comprises an input element, an output element and a coupling member, and a lock device comprising such coupling arrangement, will be described. The same or similar reference numerals will be used to denote the same or similar structural features.
- (14) FIG. 1 schematically represents a perspective view of a lock device **10**. The lock device **10** comprises a coupling arrangement **12**. The coupling arrangement **12** comprises an input element **14** rotatable about a rotation axis **16**, and an output element **18** rotatable about the rotation axis **16**. The output element **18** of this example comprises a tailpiece **20**. The tailpiece **20** is one example of a locking member according to the present disclosure.
- (15) The lock device **10** of this example further comprises a cylinder housing **22**. A part of the coupling arrangement **12** is arranged inside the cylinder housing **22**. The input element **14** protrudes from one side of the cylinder housing **22** and the tailpiece **20** is positioned on an opposite side of the cylinder housing **22**.
- (16) The lock device **10** of this example further comprises a knob **24**. The knob **24** is one example of an actuating element according to the present disclosure. The knob **24** can be rotated by hand about the rotation axis **16**. The input element **14** is fixed to the knob **24** and rotates in common with the knob **24**. The knob **24** and the input element **14** can rotate continuously around the rotation axis **16**.
- (17) The lock device **10** of this example further comprises an electric generator **26**. The electric generator **26** may for example be fixed to a door leaf (not shown). The knob **24** may in this case be rotatable relative to the door leaf.
- (18) The electric generator **26** can be driven to generate electric energy by rotation of the input element **14**. To this end, the lock device **10** of this example comprises a crown wheel **28**, a pinion **30** and a gearbox **32**. The crown wheel **28** is fixed to the input element **14** and rotates about the rotation axis **16**. The pinion **30** meshes with the crown wheel **28** and rotates about an axis perpendicular to the rotation axis **16**. The crown wheel **28** and the pinion **30** here form a bevel gear. The gearbox **32** may be a gear transmission comprising one or more intermediate gear wheels. Rotation of the pinion **30** is transmitted via the gearbox **32** to a rotation of a rotor (not shown) of the electric generator **26** to thereby generate electric energy.
- (19) FIG. 2 schematically represents a cross-sectional side view of the lock device **10**. As shown in FIG. 2, the lock device **10** further comprises an engaging member **34**, an electric motor **36** and a coupling member **38**.

(20) The input element **14** of this specific example comprises an input axle **40**, a square axle **42**, a plug **44** and a shoulder screw **46**. Each of the input axle **40**, the square axle **42**, the plug **44** and the shoulder screw **46** is concentric with, and rotatable around, the rotation axis **16**. The input axle **40** is fixed to the knob **24**. The square axle **42** is fixed to the input axle **40** by being received in the input axle **40**. The plug **44** is fixed to the square axle **42**. The shoulder screw **46** is fixed to the plug **44**. As shown in FIG. 2, the plug **44** and the shoulder screw **46** are arranged inside the cylinder housing **22**. The crown wheel **28** is fixed to the input axle **40**.

(21) The engaging member **34** is arranged inside the cylinder housing **22**. The engaging member **34** of this example comprises a cylindrical pin.

(22) The electric motor **36** is in this example arranged to be electrically powered by the electric generator **26**. The lock device **10** of this example is thus electrically powered by means of energy harvesting. The electric motor **36** may alternatively be electrically powered by a battery.

(23) The electric motor **36** comprises a threaded motor shaft **48** and an actuating member **50** threadingly engaging the motor shaft **48**. By actuating the electric motor **36** to rotate the motor shaft **48**, the actuating member **50** can be moved in a direction perpendicular to the rotation axis **16**. As shown in FIG. 2, the electric motor **36** has an elongated shape and is oriented perpendicular to the rotation axis **16**. By arranging the electric motor **36** perpendicular to the rotation axis **16**, the coupling arrangement **12** does not protrude as much from the door leaf as if the electric motor **36** would be parallel with the rotation axis **16**. The lock device **10** is thereby made more compact. The cylinder housing **22** may for example be only 13 mm long while containing the electric motor **36**. The cylinder housing **22** will thereby fit entirely into most holes bored for cylinder housings without protruding from the door face. The door face may be positioned between the electric generator **26** and the cylinder housing **22**.

(24) The coupling arrangement **12** further comprises an engaging spring **52**. The engaging spring **52** is arranged between the actuating member **50** and the engaging member **34**. The engaging spring **52** is here designed as a compression coil spring encircling the motor shaft **48**.

(25) The coupling member **38** is rotationally fixed to the input element **14**, here fixed to and surrounding the plug **44**. Thus, when the input element **14** rotates around the rotation axis **16**, also the coupling member **38** rotates around the rotation axis **16**. However, the coupling member **38** is arranged to slide relative to the input element **14** along the rotation axis **16**.

(26) The coupling member **38** of this example comprises a track **54** and a cam profile **56**. The cam profile **56** is one example of a transmission mechanism according to the present disclosure. The engaging member **34** can engage the cam profile **56** and the track **54**, as described below. The coupling member **38** further comprises a first coupling part **58**. The cam profile **56** is here arranged between the track **54** and the first coupling part **58** along the rotation axis **16**. As shown in FIG. 2, also the coupling member **38** is arranged inside the cylinder housing **22**.

(27) In addition to the tailpiece **20**, the output element **18** of this example further comprises a second coupling part **60**. The second coupling part **60** is fixed to the tailpiece **20**. The second coupling part **60** is freely rotatable relative to the shoulder screw **46**. To this end, a play is provided between the shoulder screw **46** and the second coupling part **60**. The second coupling part **60** is provided inside the cylinder housing **22** and the tailpiece **20** is provided outside the cylinder housing **22**.

(28) The coupling arrangement **12** of this example further comprises a coupling spring **62**. The coupling spring **62** is one example of a coupling force device according to the present disclosure. The coupling spring **62** acts between the the coupling member **38** and the output element **18**. The coupling spring **62** is here exemplified as a compression coil spring encircling the plug **44**.

(29) FIG. 3 schematically represents a partial perspective view of the lock device **10**. The coupling member **38** comprises a plurality of recesses **64**. In this example, the coupling member **38** comprises three recesses **64** evenly distributed around the rotation axis **16**. The track **54** is deeper than each recess **64**. The track **54** is provided around the entire circumference of the coupling

member **38** and lies in a plane perpendicular to the rotation axis **16**. In this example, an edge of each recess **64** defines one of the cam profiles **56**. Each cam profile **56** comprises two surfaces. Each such surface is here inclined approximately 60 degrees relative to the rotation axis **16**.

(30) Each of the first coupling part **58** and the second coupling part **60** has a circular circumference and is centered with respect to the rotation axis **16**. The first coupling part **58** and the second coupling part **60** form one example of a coupling according to the present disclosure. Each of the first coupling part **58** and the second coupling part **60** comprises a plurality of tapered teeth for being brought into meshing engagement. The coupling of this example is a Hirth coupling.

(31) The electric generator **26** can convert mechanical energy from a manual rotation of the knob **24** to electric energy. Electric energy harvested by manually rotating the knob **24** can thereby be used to authorize a user, to command the electric motor **36** to move the actuating member **50** from a first actuating position to a second actuating position, and to command the electric motor **36** to move the actuating member **50** from the second actuating position back to the first actuating position after some time. The electric generator **26** may function as a primary energy source for the coupling device.

(32) FIG. **4** schematically represents a block diagram of one of many implementations of the lock device **10**. The lock device **10** further comprises electronics, generally indicated by reference numeral **66**. The electronics **66** is arranged to be electrically powered by the electric generator **26**. The electronics **66** is further configured to produce an authorization signal **68** upon authorization of a user. When the authorization signal **68** is received by the electric motor **36**, the electric motor **36** drives the actuating member **50** from the first actuating position to the second actuating position.

(33) As illustrated in FIG. **4**, the electronics **66** electrically connects the electric generator **26** to the electric motor **36**. The electronics **66** of the example in FIG. **4** comprises power management electronics **70**, reading electronics **72** and credential evaluation electronics **74**. The electric generator **26**, the power management electronics **70**, the reading electronics **72**, the credential evaluation electronics **74** and the electric motor **36** are connected by electric conductors, as shown in FIG. **4**. The electric motor **36** is thereby arranged to be electrically powered by the electric generator **26**.

(34) The power management electronics **70** of this example comprises energy harvesting electronics, such as diodes (not shown) and a capacitor (not shown). The reading electronics **72** of this example comprises an antenna (not shown) for receiving an input signal, and a reading unit (not shown).

(35) When the knob **24** is manually grabbed and rotated by the hand of a user, the electric generator **26** harvests electric energy from the rotation. When sufficient electric energy has been harvested by the electric generator **26**, an authorization process is initiated. During the authorization process, the reading electronics **72** is powered by the power management electronics **70** and can for example communicate wirelessly with an external device, such as with a mobile phone via BLE (Bluetooth Low Energy). The reading electronics **72** receives a credential from the external device and sends an access signal **76**, based on the credential, to the credential evaluation electronics **74**.

(36) The credential evaluation electronics **74** then determines whether or not access should be granted based on the access signal **76**. If the authorization request is denied, the electric motor **36** is not actuated, i.e. the actuating member **50** remains in the first actuating position. If the authorization request is granted, e.g. if a valid credential is presented, the credential evaluation electronics **74** issues an authorization signal **68** to the electric motor **36**. When sufficient electric energy has been harvested by the further continuous rotation of the knob **24**, the electric motor **36** is actuated to drive the actuating member **50** from the first actuating position to the second actuating position.

(37) FIG. **5** schematically represents a perspective view of the coupling arrangement **12** where the actuating member **50** is in the first actuating position **78**, the engaging member **34** is in a first position **80**, and the coupling member **38** is in a decoupled position **82**. In the decoupled position

82 of the coupling member **38**, the first coupling part **58** is spaced from the second coupling part **60** along the rotation axis **16**. A rotation of the input element **14** is thereby not transmitted to a rotation of the output element **18**. The lock device **10** therefore cannot be unlocked (or locked) by rotation of the knob **24** when the coupling member **38** is in the decoupled position **82**.

(38) When the actuating member **50** is in the first actuating position **78**, the engaging member **34** is held in the first position **80** by the engaging spring **52**. In the first position **80**, the engaging member **34** does not engage the coupling member **38**. The coupling member **38** is thereby forced to the decoupled position **82** (to the left in FIG. 5) by means of the coupling spring **62**.

(39) FIG. 6 schematically represents a perspective view of the coupling arrangement **12** when the actuating member **50** has been driven by the electric motor **36** from the first actuating position **78** to the second actuating position **84**, e.g. in response to a granted authorization request. As shown in FIG. 6, the engaging member **34** is not aligned with any of the recesses **64**. As a consequence, the movement of the actuating member **50** from the first actuating position **78** to the second actuating position **84** causes the engaging spring **52** to be compressed. The engaging member **34** is now forced against an exterior surface of the coupling member **38**, radially outside the recesses **64** with respect to the rotation axis **16**.

(40) FIG. 6 further shows that the movement of the actuating member **50** from the first actuating position **78** to the second actuating position **84** occurs in an actuating direction **86** perpendicular to the rotation axis **16**. Thus, the actuating direction **86** is here angled 90 degrees to the rotation axis **16**.

(41) FIG. 7 schematically represents a perspective view of the coupling arrangement **12** when the engaging member **34** is in a second position **88**. By manually rotating the knob **24** when the actuating member **50** is in the second actuating position **84**, the engaging member **34** eventually becomes aligned with one of the recesses **64** in the coupling member **38**. The engaging spring **52** then forces the engaging member **34** to move radially inwards (with respect to the rotation axis **16**) from the first position **80** to the second position **88** to engage the coupling member **38**. As can be gathered from FIG. 7, a distance moved by the engaging member **34** from the first position **80** to the second position **88** is rather short and substantially corresponds to the depth of the recess **64**.

(42) Since the positioning of the actuating member **50** in the second actuating position **84** is a prerequisite for the engaging member **34** to move from the first position **80** to the second position **88**, the electric motor **36** is said to be arranged to affect movement of the engaging member **34** from the first position **80** to the second position **88**. Note that the coupling member **38** remains in the decoupled position **82** immediately after the engaging member **34** has entered the recess **64**.

(43) FIG. 8 schematically represents a perspective view of the coupling arrangement **12** when the input element **14** is rotated further by rotation of the knob **24**. As shown in FIG. 8, when the engaging member **34** is in the second position **88** and the coupling member **38** is rotated, the engagement between the engaging member **34** and the cam profile **56** causes the coupling member **38** to be pushed forward (to the right in FIG. 8) against the force of the coupling spring **62**. The coupling member **38** thereby slides forward over the input element **14** by the manual force rotating the knob **24**, and not by the force from the electric motor **36**.

(44) FIG. 9 schematically represents a perspective view of the coupling arrangement **12** when the input element **14** is rotated further. As shown in FIG. 9, further rotation of the coupling member **38** has caused the coupling member **38** to be moved further along the rotation axis **16** and into a coupled position **90**. At the same time, the input element **14** becomes aligned with the track **54**. The engaging spring **52** thereby forces the engaging member **34** to move radially inwards (with respect to the rotation axis **16**) from the second position **88** to a third position **92** engaging the track **54**. As can be gathered from FIG. 9, a distance moved by the engaging member **34** from the second position **88** to the third position **92** is rather short and substantially corresponds to a radial distance (with respect to the rotation axis **16**) between a bottom of the track **54** and a bottom of the recess **64**.

(45) By means of the engagement between the engaging member **34** and the cam profile **56** when the engaging member **34** is in the second position **88**, a rotation of the coupling member **38** about the rotation axis **16** is transmitted to a movement of the coupling member **38** from the decoupled position **82** to the coupled position **90**. In this way, the transmission mechanism, here exemplified by the cam profiles **56**, is arranged to transmit a rotation of the coupling member **38** about the rotation axis **16** to a movement of the coupling member **38** along the rotation axis **16** from the decoupled position **82** to the coupled position **90** when the engaging member **34** is in the second position **88**.

(46) When the coupling member **38** is in the coupled position **90**, the input element **14** and the output element **18** rotate in common by the engagement between the first coupling part **58** and the second coupling part **60**. The coupling member **38** is held in the coupled position **90** (by the engaging member **34** being in the third position **92**) as long as the actuating member is in the second actuating position **84**. Since the positioning of the actuating member **50** in the second actuating position **84** is a prerequisite for the engaging member **34** to move from the second position **88** to the third position **92**, the electric motor **36** is said to be arranged to affect movement of the engaging member **34** from the second position **88** to the third position **92**.

(47) FIG. **10** schematically represents a perspective view of the coupling arrangement **12** when the output element **18** is rotated by rotation of the input element **14**. The rotation of the tailpiece **20** causes unlocking of the lock device **10**, for example by driving a bolt in a lock case (not shown).

(48) The knob **24** can be continuously rotated about the rotation axis **16** during the authorization procedure. Electric energy harvested by manually rotating the knob **24** can thereby be used to authorize a user and to actuate the electric motor **36** to drive the actuating member **50** from the first actuating position **78** to the second actuating position **84**. The user can rotate the knob **24** continuously during the authorization process, the subsequent switching process of the actuating member **50**, and the subsequent rotation of the tailpiece **20**. Thereby, a seamless access is provided.

(49) FIG. **11** schematically represents a perspective view of the coupling arrangement **12** when the actuating member **50** has returned to the first actuating position **78**. When the actuating member **50** returns to the first actuating position **78**, the engaging spring **52** pulls the engaging member **34** from the third position **92** out from the track **54** and back to the first position **80**. The coupling spring **62** then pushes the coupling member **38** from the coupled position **90** back to the decoupled position **82**.

(50) While the present disclosure has been described with reference to exemplary embodiments, it will be appreciated that the present invention is not limited to what has been described above. For example, it will be appreciated that the dimensions of the parts may be varied as needed. Accordingly, it is intended that the present invention may be limited only by the scope of the claims appended hereto.

Claims

1. A coupling arrangement for a lock device, the coupling arrangement comprising: an input element rotatable about a rotation axis; an output element rotatable about the rotation axis; an engaging member movable between a first position and a second position; an electric motor arranged to affect movement of the engaging member between the first position and the second position; a coupling member rotationally locked to the input element with respect to the rotation axis and axially movable relative to the input element along the rotation axis between a decoupled position, in which a rotation of the input element is not transmitted by the coupling member to a rotation of the output element, and a coupled position, in which a rotation of the input element is transmitted by the coupling member to a rotation of the output element, wherein the coupling member comprises a circular track, and wherein the engaging member is arranged to engage the track; and a transmission mechanism arranged to transmit a rotation of the coupling member to the

axial movement of the coupling member from the decoupled position to the coupled position when the engaging member moves to the second position; wherein the electric motor is arranged to affect movement of the engaging member from the second position to a third position, wherein the second position lies between the first position and the third position, and wherein the engaging member is arranged to engage the track when the engaging member moves to the third position and the coupling member axially moves to the coupled position.

2. The coupling arrangement according to claim 1, wherein the electric motor is arranged to affect linear movement of the engaging member in an actuating direction between the first position and the second position.

3. The coupling arrangement according to claim 2, wherein the actuating direction and the rotation axis are non-parallel.

4. The coupling arrangement according to claim 2, wherein the actuating direction is angled approximately 90 degrees to the rotation axis.

5. The coupling arrangement according to claim 1, further comprising an actuating member arranged to be driven by the electric motor, and an engaging spring arranged to force the engaging member to move relative to the actuating member from the first position towards the second position.

6. The coupling arrangement according to claim 5, wherein the actuating member is movable between a first actuating position and a second actuating position, and wherein the engaging spring is arranged to force the engaging member to move from the first position to the second position when the actuating member moves to the second actuating position.

7. The coupling arrangement according to claim 6, wherein the engaging spring is arranged to force the engaging member to move from the second position to the third position when the actuating member moves to the second actuating position.

8. The coupling arrangement according to claim 1, wherein the transmission mechanism comprises at least one cam profile, and wherein the engaging member is arranged to engage one of the at least one cam profile in the second position.

9. The coupling arrangement according to claim 1, further comprising a coupling arranged to transmit the rotation of the coupling member to the rotation of the output element when the coupling member axially moves to the coupled position.

10. The coupling arrangement according to claim 9, wherein the coupling is a Hirth coupling.

11. The coupling arrangement according to claim 1, further comprising a coupling force device arranged to force the coupling member to axially move towards the decoupled position.

12. A lock device comprising a coupling arrangement according to claim 1.

13. The lock device according to claim 12, further comprising a cylinder housing, wherein the electric motor, the engaging member, and the coupling member are arranged inside the cylinder housing.

14. A coupling arrangement for a lock device, the coupling arrangement comprising: an input element rotatable about a rotation axis; an output element rotatable about the rotation axis; an engaging member movable between a first position and a second position; an electric motor arranged to affect movement of the engaging member between the first position and the second position; an actuating member arranged to be driven by the electric motor, and an engaging spring arranged to force the engaging member to move relative to the actuating member from the first position towards the second position, wherein the actuating member is movable between a first actuating position and a second actuating position, and wherein the engaging spring is arranged to force the engaging member to move from the first position to the second position when the actuating member moves to the second actuating position; a coupling member rotationally locked to the input element with respect to the rotation axis and axially movable relative to the input element along the rotation axis between a decoupled position, in which a rotation of the input element is not transmitted by the coupling member to a rotation of the output element, and a

coupled position, in which a rotation of the input element is transmitted by the coupling member to a rotation of the output element, the coupling member comprising a circular track; and a transmission mechanism arranged to transmit a rotation of the coupling member to the axial movement of the coupling member from the decoupled position to the coupled position when the engaging member moves to the second position; wherein the electric motor is arranged to affect movement of the engaging member from the second position to a third position, wherein the second position lies between the first position and the third position; and wherein the engaging member is arranged to engage the track when the engaging member moves to the third position and the coupling member axially moves to the coupled position.

15. The coupling arrangement according to claim 14, wherein the electric motor is arranged to affect linear movement of the engaging member in an actuating direction between the first position and the second position.

16. The coupling arrangement according to claim 15, wherein the actuating direction and the rotation axis are non-parallel.

17. The coupling arrangement according to claim 14, wherein the transmission mechanism comprises at least one cam profile, and wherein the engaging member is arranged to engage one of the at least one cam profile in the second position.

18. The coupling arrangement according to claim 14, wherein the engaging spring is arranged to force the engaging member to move from the second position to the third position when the actuating member moves to the second actuating position.

19. The coupling arrangement according to claim 14, further comprising a coupling arranged to transmit the rotation of the coupling member to the rotation of the output element when the coupling member axially moves to the coupled position.
